INCH-POUND

A-A-55748
March 27, 1995

COMMERCIAL ITEM DESCRIPTION

SPRAYERS, INSECTICIDE, MANUALLY-CARRIED: HAND-OPERATED-COMPRESSION

The General Services Administration has authorized the use of this commercial item description as a replacement for Military Specification MIL-S-14102, which is cancelled.

1. SCOPE. This commercial item description covers manually carried, hand-operated-compression insecticide sprayers. The sprayers are intended for manual spraying of insecticides and fungicides.

2. CLASSIFICATION. The sprayers shall conform to the following sizes. Operating capacity is defined as maximum liquid fill of the tank when first used - 3/4 liquid fill arid 1/4 air capacity with all accessories in place:

Size 1 - 1 gallon operating capacity.

Size 2 - 2 gallon operating capacity.

3. SALIENT CHARACTERISTICS.

3.1 Description The sprayers shall be of the cylindrical type and equipped with a hand-operated air pump for pressurization, hose, cut-off valve, lance, nozzle and other accessories as specified herein.

Beneficial connents, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Defense Construction Supply Center, P.O. Box 3990, ATTN: DCSC-VSS, Columbus, Ohio 43216-5000.

AMSC N/A FSC 3740

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3.2 Materials. The entire tank, including the removable filler-hole cover, shall be made of stainless steel conforming to ASTM A 167, UNS S30200 or S30400 or made of forged brass. The springs shall be made of stainless steel conforming to ASTM A 313, UNS S30200 or better. All other parts that normally come in contact with the insecticide shall be of the same type of stainless steel specified above, or of brass, bronze, or copper alloy. Joining materials or welds and heat-affected zones shall be corrosion resistant. All tubing shall be stress corrosion resistant.

3.3 Dimensions The maximum height of the tank from the bottom of the base to the uppermost point of the top shall not exceed 11 inches (27.9 cm) for size 1 and 22 inches (55.9 cm) for size 2. The height from the base of the sprayer to the handle in its lowest position shall not exceed 6 additional inches (15.2 cm) above the height of the tank. The diameter of the tank shall not exceed 9 inches (22.9 cm).

3.4 Weight. The weight of the sprayer, when empty but equipped for operation, shall not exceed 8 pounds (3.6 kg) for size 1 and 10 pounds (4.5 kg) for size 2.

3.5 Tank markings The tank shall be marked in a legible and permanent manner with horizontally embossed lines to indicate the fill charges. The fill marks shall appear at 0.5 gallon (2L) intervals. The maximum fill charge shall be at 3/4 of the total tank capacity. Other markings shall indicate the year of manufacture.

3.6 Tank fittings.

3.6.1 Fastenings for lance The tank shall be fitted with a pair of fastenings to secure the nozzle and lance and when specified (see 7.2.3), extra fastenings for an extension lance. Fastenings shall withstand dropping to the extent that they may be bent back to their original shape and remain functional. No set screws shall be used for fastenings.

3.6.2 Filler hole. The filler hole shall be not less than 3.7 inches nominal (94 mm) in diameter, if circular, or in minor axis, if oval. It shall have no sharp edges. The filler hole cover shall be attached t-o the tank with a chain capable of withstanding a breaking strain of not less than 110 pounds (50 kg).

3.6.3 Riveted fittings Rivets used for securing lance brackets or other external fittings shall not penetrate the pressurized portion of the tank.

3.6.4 Pressure-relief (safety) valve The tank shall be equipped with a fixed pressure-relief valve for whenever an air-inlet valve is used to prevent pressurization in the tank beyond the limits prescribed for safety or operational purposes. Threaded fittings subject to pressure shall be channeled.

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3.6.5 Plug for pressure gauge opening A plug shall be provided for the opening into which a pressure gauge can fit to provide an air-tight and liquid-tight closure, should the gauge not be required (see notes).

3.7 Pump.

3.7.1 Construction. The pump cylinder shall be of seamless or electrically welded construction if made-of stainless steel or brass. The pump cylinder shall be a separate part, independently detachable from the tank, tank top, or any other component of the sprayer. The upper part of the cylinder shall be cone-shaped to facilitate reassembly of the plunger cup.

3.7.2 Check valve The pump check valve shall be of synthetic rubber such as polysulfide (i.e., Thiokol) or fluorocarbon (i.e., Viton), shore hardness 67-73, or equal, and shall be air-tight and liquid-tight. It shall be so designed as to prevent any leaking of liquid into the pump cylinder over a period of 15 minutes when the tank is pressurized between 65-70 psi (447.8 - 482.2 Kpa) and the check valve is fully submerged.

3.7.3 Plunger shaft The plunger shaft shall be of steel and may be either a solid rod or a tube.

3.7.4 Plunger cup. The plunger cup shall be not less than 0.06 inch (1.5 mm) thick and the cup skirt shall be not less than 0.5 inch (13 mm) in length. It shall be fungus resistant.

3.7.5 Plunger cup assembly. The plunger cup assembly shall be so constructed that whenever positive downward pressure is applied to the plunger rod when in the raised position, the plunger cup skirt is pressed against the pump cylinder wall to form a seal and provide pressurization.

3.7.6 Handle. The pump handle shall be of the D-type and shall have a grip of not less than 3.56 inches (9 mm) inside the grip support. The pump handle shall be so designed that a complete stroke of the pump may be obtained with the pump handle turned to any position. The compression strength of the assembly, including the handle, its components, and its junction with the plunger rod, shall not be less than that of the plunger rod. Whatever the material of manufacture, it shall be nonabsorbent.

3.7.7 **Handle locking device.** The plunger shaft shall befitted with a quick-acting locking device that, when applied, will secure the plunger assembly nearest to the lowest operating position for carrying.

3.8 Dip tube. The dip tube shall be constructed of brass, stainless steel, or other suitable material, and shall extend to within approximately 0.4 inch (10.2 mm) of the bottom of the tank. The end of the tube within the tank shall be cut at an angle to prevent blockage. The dip tube shall be securely supported at the point where it emerges from the tank.

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3.9 Strainers. Two strainers shall be provided. The two strainers shall be of equal dimensions located at different points easily accessible in the discharge line between the entry to the dip tube and the entry ports of the cut-off valve. One strainer shall be provided in the cut-off valve. The strainers shall be made of stainless steel, bronze, copper alloy, or any material of equal or superior durability and corrosion resistance. The strainers shall have sufficient structural strength or shall be mounted in such a way that they will not be subject to accidental puncture or collapse. They shall be easily removable and replaceable without special tools, and it is preferable that they be pushed into place rather than screwed. Flow through the strainers shall be from the outer surface inward. The strainers shall allow easy passage of liquid.

3.10 Hose. Unless otherwise specified (see 7.2.4), hose shall be of polychloroprene rubber, polyvinyl chloride or similar material and shall be resistant to pesticide formulations or oils. A length 5 feet (1.5 m) with an inside diameter between 0.25 to 0.375 inch (6.4 to 9.5 mm) shall be provided with each sprayer. The hose shall have one or more plies of reinforcement made from cotton duck, cotton yarn, or synthetic fiber, of at least two-end yarn. The material shall be evenly and firmly woven and free from unsightly defects such as dirt, knots and lumps, as well as from irregularities of twist. The tube and cover shall be free from pitting and shall be uniform in thickness. The reinforcement shall be lapped (not sewn) at least 0.5 inch (13 mm) and shall be well frictioned on both sides with a rubber compound that shall firmly join the plies to the rubber tube and cover and to each other.

3.11 Hose connections. Hoses shall be retained on connections or couplings preferably by the worm-drive clamp. The connections shall have no sharp edges and a minimum of extended projections liable to catch on clothing or other objects. Threaded terminal connections shall have finished, hexagonal faces or opposing flattened surfaces. Nuts of adequate strength and size may be permitted if couplings are fabricated to enable air-tight and liquid-tight joints to be made by thumb pressure at the highest designated operating pressure.

3.12 Cut-off valve.

3.12.1 Construction. All metal parts of the valve body shall be made of brass, bronze, anodized aluminum, or stainless steel. The valves shall be so constructed that inner and outer parts are readily accessible for frequent cleaning without the use of special tools.

3.12.2 Connections. Valves shall be equipped with screw threads at inlet and outlet ports. The outer surfaces of the valve, adjacent to the screw threads, shall have opposing faces, to permit the use of a wrench, if tools are required for its removal. Knurled faces or wing nuts of adequate shape

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and size may be used provided that the valve can be loosened and tightened adequately by the pressure of thumb and fingers alone, without the use of tools, at the highest designated operating pressure.

3.12.3 Valve handle. The handle shall be conveniently located on the valve to permit actuation by hand. The valve handle and all associated parts shall conveniently fit the operator's hand without discomfort, and shall have. no sharp edges or projections. The pressure required to move the handle on a lever-type valve from the closed to the open position shall be not more than 30 pounds per square inch (207 Kpa). The length of the valve handle shall not be less than 3.5 inches (9 cm) measured from the pivot.

3.12.4 Leakage. Cut-off valves shall not drip or leak through the valve or around the packing when subjected to normal operating pressures over a period of one year.

3.13 Lance. The lance shall be brass or stainless steel and of seamless construction and shall be easily detachable. It shall have the same thread and size at both ends so that any number of lances in series, or the nozzle alone, may be connected directly to the cut-off valve housing. Joints shall be leakproof and body facings shall have hexagonal or flattened faces. Nuts of adequate shape and size may be used if couplings are fabricated to enable air-tight and liquid-tight joints to be made by thumb pressure at the highest designated operating pressure. When specified, (see 7.2.5), a gasket shall be required to ensure a tight fit. Standard lances shall be between 16 and 20 inches (40-50 cm) in length, measured between the external faces of the joints.

3.14 Nozzle.

3.14.1 Description The spray nozzle shall be of the hydraulic-energy type in which droplets are formed by forcing the spray material through an orifice. It shall consist of a droplet-producing nozzle tip plus a nozzle body and nozzle cap. The nozzle tip shall be removable and replaceable without the use of tools, and shall not leak after reassembly. When specified (see 7.2.6), gaskets of polyethylene or equivalent materials shall be required.

3.14.2 Nozzle assembly. The nozzle assembly consisting of the body, cap and tip shall be constructed from stainless steel, ceramic or brass.

3.15 Gaskets. All gaskets shall be capable of being readily fitted to the sprayer. Where possible, gaskets shall be wholly or partially recessed into the parts into which the joints are made. Polyethylene or gaskets of other suitable material may be used for working with water suspensions.

3.16 Optional items. When required, the following items shall be specified (see 7.2.7):

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3.16.1 Air-inlet valve. An air-inlet valve (automobile type) for the purpose of supplying air pressure to the sprayer from an outside pump or high-pressure source. With such an inlet valve, a highly reliable pressure/relief (safety) valve shall be required (see 3.6.4).

3.16.2 Regulated delivery system. A regulated flow-delivery system, consisting of a complete nozzle assembly and disc flow regulator. The discharge rate and distribution pattern shall be as specified (see 7.2.8).

3.16.3 Constant-pressure regulator. The sprayer may be equipped with a constant-pressure regulator inserted in the discharge line, adjustable to produce tip pressures in the program operating range, as specified (see 7.2.9).

3.16.4 Lances. Extra lances for use as extensions. These may be the same length as furnished with the sprayer or as specified (see 7.2.10). Leakproof telescopic lances may be provided (see 7.2.11).

3.16.5 Swivel nozzles. Leakproof swivel nozzles that can readily be rotated by hand. Additional swivel nozzles shall be as specified (see 7.2.12).

3.16.6 Multiple spray nozzle. A multiple spray nozzle which has an adjustable cone spray and straight stream application.

3.16.7 Gooseneck attachment. A gooseneck attachment consisting of a section of lance, curved 120 degrees, with 8 inches (20cm) length of tubing, with a coupling on each end to permit attachment to the lance nozzle body.

3.16.8 Gooseneck lance. A nine inch gooseneck lance with an extra 20 inch (50.8 cm) extension lance. Additional fastenings will be required on the tank for the attachment of the lance. A clip for the lance will not be required.

3.16.9 Nylon nozzle body. A nozzle body and cap made of shockproof nylon (containing not less than 2 percent moisture by weight) in place of the normal brass fitting.

3.16.10 Nozzle gaskets. Gaskets of polyethylene, or equivalent material, in such places and number in the nozzle body as required to make that unit easily and quickly disassembled and assembled without tools. The unit assembled in such a manner shall not leak when there is flow through the nozzle.

3.16.11 Pressure gauge. A back-mounting pressure gauge, preferably of the diaphragm type, with a scale reading from 0 to 101.5 psi (O to 700 Kpa). The body material shall be brass, bronze, or other metal or material having equal or superior corrosion resistance and strength. The sensing element chamber shall be "non-pocketing" to prevent interference with the proper operation of

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the sensing element by the deposition and accumulation of spray materials. In the case of diaphragm type, the inlet port shall have an easily removable filter assembly, or other means to prevent the entry of suspended material. The dial cover shall be shatterproof and moisture-proof and permit easy reading of the tank pressure.

3.16.12 Foot-rest. A foot-rest designed to steady the apparatus while it is pumped. It shall be attached to the bottom part of the tank and shall be hinged to fold up against the tank side and remain there until required.

3.16.13 Inspection light. A spray inspection light. It shall be attached to the lance and be constructed from brass, aluminum or plastic.

3.16.14 **Crack and crevice tip assembly.** A crack and crevice nozzle tip assembly. Component; shall be made of stainless steel, ceramic or a suitable nonferrous material.

3.16.15 Treatment and fresh Unless otherwise specified (see 7.2.13), the sprayer shall be treated and finished in accordance with the quality practiced by the manufacturer to obtain treatment and finish of equivalent commercial units.

3.17 Identification marking.

3.17.1 Civil identification marking. The sprayer shall be marked in a permanent and legible manner with the manufacturer’s name or with a trademark of such known character that the source of manufacture may be readily determined.

3.17.2 Military identification marking. The sprayer shall be identified in accordance with MIL-STD-130.

3.18 Workmanship. All parts, components, and assemblies of the sprayer shall be clean and free from harmful extraneous material such as sand, dirt, fins, pits, scale, and rust, and shall also be free from burrs and sharp edges or projections or other defects that could impair its performance or service-ability.

3.19 Operator's manual. The manufacturer’s standard instruction manual with parts lists shall be provided with each sprayer.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. QUALITY ASSURANCE PROVISIONS.

5.1 Contractor certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING.

6.1 Preservation. packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Purchase approval The purchaser should obtain Major Command pest management professional consultant approval before purchase.

7.2 Ordering data. The purchaser should specify the following:

7.2.1 Number, title, and date of this commercial item description.
7.2.2 Size of sprayer required.
7.2.3 When extra fastenings will be required for extension lance (see 3.6.1).
7.2.4 When hose material other than as specified is required (see 3.10).
7.2.5 When a gasket will be required to ensure tight fit of lance (see 3.13).
7.2.6 When gaskets of polyethylene or equivalent materials will be required with nozzle (see 3.14.1).
7.2.7 Optional items to be included with sprayer (see 3.16).
7.2.8 The discharge rate and distribution pattern when a regulated delivery system is required (see 3.16.2).
7.2.9 When a constant-pressure regulator is required (see 3.16.3).
7.2.10 The number and length of extra lances required (see 3.16.4).
7.2.11 When a leakproof telescopic lance is required (see 3.16.4).
7.2.12 When additional swivel nozzles are required (see 3.16.5).
7.2.13 When treatment and finish will be other than the manufacturer's equivalent commercial units (see 3.16.15).

7.3 Source of Documents.

7.3.1 ASTM A 167 and A 313 may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.

7.3.2 MIL-STD-130 may be obtained from the Defense Printing Service Detachment Office, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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7.4 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID.

NSN	Size
3740-00-191-3677	1
3740-00-641-4719	2

MILITARY INTERESTS:

Custodians:
Air Force - 99
Navy - YD-l

Review Activities:
Army - MD-1, MS
Air Force - 84
Navy - CG, MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

DLA-CS

Project Number 3740-0137

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